

QUANTUM – ASSISTED MACHINE LEARNING FOR BRAIN COMPUTER INTERFACE SYSTEMS

A MINI PROJECT REPORT

Submitted by

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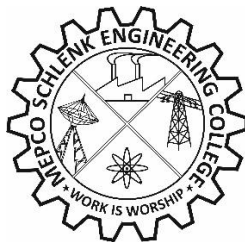
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BONAFIDE CERTIFICATE

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ABSTRACT

Brain–Computer Interface (BCI) systems are rapidly developing technologies that enable direct communication between the human brain and external devices using neural signals such as Electroencephalography (EEG). These systems play an important role in areas like healthcare, assistive technologies, and human–computer interaction. However, accurate classification of EEG signals remains a challenging task due to their high dimensionality, nonlinear characteristics, sensitivity to noise, and variations between individuals. To overcome these challenges, this project presents a comparative machine learning framework for EEG signal classification using three different approaches: Support Vector Machine (SVM), Convolutional Neural Network (CNN), and Variational Quantum Neural Network (VQNN). Initially, the EEG Eye State dataset is pre-processed using normalization to standardize feature values and improve data quality. Principal Component Analysis (PCA) is then applied to reduce the dimensionality of the data while retaining the most important features, which are used for SVM classification.

In addition, a CNN model is implemented to automatically learn spatial patterns from the EEG data by reshaping it into a structured format, reducing the need for manual feature extraction. Furthermore, a VQNN model is developed using parameterized quantum circuits, where classical data is encoded into quantum states to capture complex nonlinear relationships in EEG signals. The performance of all three models is evaluated and compared using accuracy as the primary metric. For SVM, an accuracy of 0.67 is noted, for cnn, an accuracy of 0.69 is noted and for qnn the accuracy has increased to 0.71. The results provide valuable insights into the strengths of classical and quantum-based approaches and demonstrate that advanced learning techniques can improve EEG signal classification. Overall, this work contributes to the development of more reliable, efficient, and intelligent BCI systems for real-world applications.

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SDGs ALIGNMENT

SDG Icons	Relevance of SDG in the context of project	Section Name(s)	Page number(s)
	This project supports better healthcare by using EEG signals to help in understanding brain activity. It can contribute to assistive systems for neurological conditions and improve the way brain-related data is analyzed for medical support.	Introduction, Problem Description, Dataset Description	1, 2, 13
	The work introduces a quantum-based approach using QNN, reflecting innovation in machine learning and quantum computing. It supports the development of advanced models suitable for near-term quantum (NISQ) systems.	System Design, System Implementation, Future Enhancements	11, 18, 27
	This project also contributes to learning and research in emerging technologies like quantum machine learning and EEG signal processing. It helps build understanding of advanced AI techniques in academic environments.	Literature Survey, Future Enhancements, Conclusion	4, 27

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LIST OF ABBREVIATIONS

ABBREVIATION	DESCRIPTION
BCI	Brain Computer Interface
COBYLA	Constrained Optimization BY Linear Approximations
EEG	Electro Encephalo Gram
EEGMM	Electro Encephalo Gram M otor M ovement
NISQ	Noisy Intermediate Scale Quantum
PCA	Principal Component Analysis
QC	Quantum Computing
QML	Quantum Machine Learning
QNN	Quantum Neural Network
SNR	Signal-to-Noise-Ratio
VQC	Variational Quantum Circuit

CHAPTER 1

INTRODUCTION

1.1 PROBLEM DESCRIPTION

Brain Brain–Computer Interface (BCI) systems are designed to build a direct connection between the human brain and external devices, allowing communication without the need for physical movement or muscle activity. This technology is especially useful for individuals with motor disabilities and is also widely applied in neurorehabilitation, cognitive monitoring, and advanced human–computer interaction. Among the different techniques available, Electroencephalography (EEG) is one of the most commonly used methods because it is non-invasive, affordable, and suitable for real-time applications. EEG works by capturing electrical signals from the brain and converting them into meaningful information that machines can understand.

Even with these advantages, EEG-based BCI systems still face several real-world challenges. The signals are usually weak and can easily be affected by noise from eye movements, muscle activity, and external disturbances. Another difficulty is that EEG signals differ from person to person, which makes it challenging to develop a model that performs consistently for all users. In addition, brain signals are highly complex and nonlinear, making accurate classification more difficult.

Traditional machine learning methods such as Support Vector Machines (SVM), Logistic Regression, and Artificial Neural Networks have been widely used for EEG signal classification. Although these approaches can produce reasonable results, they often struggle to capture deeper and more complex patterns in the data. They also require significant preprocessing and manual feature selection, which can increase complexity and reduce efficiency, especially when dealing with high-dimensional EEG data.

To address these limitations, this project follows a comparative approach using three models: PCA-based Support Vector Machine (SVM), Convolutional Neural Network (CNN), and Variational Quantum Neural Network (VQNN). First, the EEG data from the Eye State dataset is pre-processed and standardized to improve its quality. Then, Principal Component Analysis (PCA) is applied to reduce the number of features while retaining the most important information. These processed features are used as input for the SVM model.

In addition, a CNN model is implemented to automatically learn spatial patterns from the EEG data by reshaping it into a structured format. This helps in extracting important features

without the need for manual intervention. Furthermore, a Variational Quantum Neural Network (VQNN) is developed using parameterized quantum circuits, where classical data is encoded into quantum states to capture complex nonlinear relationships more effectively.

The performance of all three models—SVM, CNN, and VQNN—is evaluated using standard metrics such as accuracy. By comparing these models, the project aims to identify the most suitable approach for EEG signal classification. This analysis also helps in understanding how classical and quantum-based methods perform on real EEG data and highlights the potential of quantum techniques in improving the performance of BCI systems.

1.2 PURPOSE OF THE PROJECT

The purpose of this project is to improve how Brain–Computer Interface (BCI) systems understand and classify EEG signals with greater accuracy and reliability. In real-world situations, EEG data is often noisy, complex, and varies from person to person, which makes it difficult for traditional machine learning models to perform consistently. These challenges can reduce the effectiveness of BCI systems, especially in areas like assistive technology, healthcare, and human–computer interaction.

To overcome these issues, this project follows a comparative approach using three different models: Support Vector Machine (SVM), Convolutional Neural Network (CNN), and Variational Quantum Neural Network (VQNN). First, the EEG data is pre-processed and standardized to improve its quality. Then, Principal Component Analysis (PCA) is applied to reduce the number of features while keeping the most important information. These refined features are used for classification with the SVM model.

In addition, a CNN model is used to automatically learn patterns from the EEG data by capturing spatial relationships, which reduces the need for manual feature extraction. The project also explores a VQNN model that applies quantum-inspired techniques to learn complex and nonlinear patterns that are difficult for classical models to capture.

Another important goal of this work is to evaluate and compare how these models perform under the same conditions. By analysing their accuracy and overall behaviour, the project aims to determine which approach is more effective for EEG signal classification. Overall, this study helps in understanding the performance of both classical and quantum-based methods and supports the development of more reliable and efficient BCI systems for real-world applications.

1.3 OBJECTIVE OF THE WORK

The objective of the project is,

- To develop an efficient framework for classifying EEG signals in Brain–Computer Interface (BCI) systems using quantum-assisted machine learning techniques.
- To analyze EEG data and perform preprocessing steps such as normalization, noise handling, and dimensionality reduction.
- To reduce the dimensionality of EEG data using Principal Component Analysis (PCA) for better efficiency and performance.
- To implement a Support Vector Machine (SVM) model for classification using PCA-reduced features.
- To design a Convolutional Neural Network (CNN) model to automatically learn spatial patterns from EEG data.
- To develop a Variational Quantum Neural Network (VQNN) to capture complex nonlinear relationships in EEG signals.
- To compare the performance of SVM, CNN, and VQNN models to achieve better accuracy, robustness, and generalization.
- To evaluate and visualize system performance using standard metrics for effective analysis and decision-making.

1.4 OUTCOMES OF THE WORK

The outcomes of the project are,

- The proposed comparative approach successfully classifies EEG signals using SVM, CNN, and VQNN models with improved accuracy.
- The preprocessing and PCA-based dimensionality reduction help in effectively handling high-dimensional EEG data and improve overall performance.
- The SVM model provides stable and efficient classification using optimized features from PCA.
- The CNN model effectively learns spatial patterns from EEG data, reducing the need for manual feature extraction.
- The VQNN model demonstrates the ability to capture complex nonlinear relationships in EEG signals using quantum-inspired techniques.
- The system provides better understanding and visualization of model performance, supporting effective analysis and decision-making.

CHAPTER 2

LITERATURE SURVEY

2.1 QUANTUM-BASED MACHINE LEARNING FOR AUTISM DETECTION USING EEG SIGNALS

Technique of Work Use

Saranya et al. (2025) proposed a quantum-based machine learning framework for detecting autism spectrum disorder using EEG signals. The methodology involves preprocessing raw EEG data to remove noise and artifacts, followed by feature extraction using Common Spatial Pattern (CSP), which enhances discriminative spatial features. These extracted features are then fed into a Quantum Neural Network (QNN), which utilizes quantum kernel mapping to transform classical data into a higher-dimensional Hilbert space. This transformation allows better separation of complex EEG patterns that are difficult to distinguish using classical methods.

Pros

One of the major advantages of this approach is its ability to achieve high classification accuracy with a reduced number of EEG channels, which simplifies data acquisition. The use of QNN enables nonlinear feature representation through parameterized quantum circuits and data encoding in Hilbert space, allowing effective modeling of complex EEG patterns. Additionally, the system demonstrates improved computational efficiency in feature representation compared to traditional machine learning techniques.

Limitations

Despite its effectiveness, the model is tested on relatively small datasets, which limits its generalization to diverse populations. The dependency on quantum simulation environments also increases computational overhead. Furthermore, real-time implementation remains challenging due to hardware limitations.

2.2 DEEP EEG FEATURE EXTRACTION USING WAVELET TRANSFORM WITH QNN AND SVM

Technique of Work Use

Tudisco et al. (2025) introduced a hybrid framework combining classical and quantum methods has been introduced for EEG classification. In this approach, the wavelet transform is used to break down EEG signals into multiple frequency components, allowing effective extraction of

time-frequency features. These features are then analyzed using both Support Vector Machines (SVM) and Quantum Neural Networks (QNN) to compare their performance and enhance classification results.

Pros

Wavelet-based feature extraction improves signal representation, making it easier to capture non-stationary and transient characteristics of EEG data. The combination of classical and quantum models increases flexibility and overall robustness. It also improves the system's ability to handle noise effectively.

Limitations

QNN models may suffer from overfitting, especially when trained with limited data. The need for parameter tuning adds complexity, and computational cost is higher due to the use of quantum simulations.

2.3 VARIATIONAL QUANTUM NEURAL NETWORK (VQNN) FOR EEG CLASSIFICATION

Technique of Work Use

A Variational Quantum Neural Network (VQNN) model has been proposed for EEG signal classification using quantum computing principles. In this approach, preprocessed EEG features are encoded into quantum states using rotation gates. The quantum circuit is designed with entanglement layers, such as CNOT gates, to capture relationships between different features. Trainable parameters are introduced through parameterized rotation gates, allowing the model to learn patterns in the data. The output is obtained by measuring expectation values from the quantum circuit, which are then used for classification through a final classical layer.

Pros

The VQNN model is effective in learning complex and nonlinear patterns present in EEG signals. It provides better feature representation compared to traditional models and can handle high-dimensional data more efficiently. The use of quantum circuits also enhances the model's ability to capture subtle variations in brain signals.

Limitations

However, the model requires significant computational resources due to quantum simulation. Training can be slow and sensitive to parameter initialization. Additionally, the performance depends on the availability and stability of quantum hardware, which may limit practical implementation in real-time applications.

2.4 QUANTUM NEUROMORPHIC COMPUTING FOR EEG SIGNAL CLASSIFICATION

Technique of Work Use

A quantum neuromorphic computing model has been introduced by combining quantum computing concepts with spiking neural networks. This system uses quantum spiking neurons that imitate biological neural behavior to process EEG signals over time. EEG data is encoded into quantum states, and quantum operations are applied to capture both spatial and temporal relationships in brain signals.

Pros

The model effectively captures both spatial and temporal features of EEG data, which are important for accurate classification. It also offers a biologically inspired learning mechanism that improves pattern recognition. The use of quantum computing further enhances the ability to process complex neural data.

Limitations

The approach requires specialized quantum hardware and advanced implementation techniques, making real-world deployment difficult. The integration of quantum and neuromorphic components also increases system complexity.

2.5 QUANTUM CNN-LSTM MODEL FOR EEG-BASED APPLICATIONS

Technique of Work Use

A hybrid Quantum CNN-LSTM model has been proposed by combining convolutional neural networks, long short-term memory networks, and quantum computing. The CNN extracts spatial features from EEG signals, while the LSTM captures temporal dependencies. Quantum layers are added to improve feature representation and overall classification performance.

Pros

This model achieves high accuracy by effectively learning both spatial and temporal characteristics of EEG signals. It also supports real-time applications due to faster processing and lower latency. The inclusion of quantum techniques further strengthens learning capability.

Limitations

The model increases computational complexity and requires significant resources for training. The use of quantum components also adds to the complexity of system design and reduces accessibility.

2.6 COMPARATIVE STUDY OF EEG SIGNAL PROCESSING TECHNIQUES

Technique of Work Use

A detailed comparative study has been carried out to evaluate different EEG preprocessing and classification methods. Various filtering techniques, feature extraction approaches, and machine learning models are analyzed to identify the most effective combinations for EEG classification tasks.

Pros

The study provides clear insights into the strengths and weaknesses of different techniques. It helps in selecting suitable preprocessing methods that improve classification accuracy and robustness. It also serves as a useful reference for choosing appropriate models for EEG applications.

Limitations

The study mainly focuses on classical approaches and does not consider quantum techniques. Therefore, it does not fully explore the benefits of quantum machine learning for complex EEG data.

2.7 EEG-BASED USER ENGAGEMENT ANALYSIS

Technique of Work Use

An EEG-based system has been designed to analyze user engagement levels using machine learning methods. The system processes EEG signals to detect patterns related to attention and cognitive activity, allowing continuous monitoring in real-time applications.

Pros

The system supports real-time interaction and provides useful insights into user behavior. It helps in developing adaptive systems that respond based on user engagement. This improves interaction quality in areas like education, gaming, and virtual environments.

Limitations

The accuracy of this system is lower compared to advanced quantum-based models. It is also affected by noise and variability in EEG signals, which can reduce reliability.

2.8 VARIATIONAL QUANTUM CIRCUITS FOR EEG CLASSIFICATION

Technique of Work Use

Variational quantum circuits have been applied for EEG classification by combining classical preprocessing with quantum circuit-based learning. EEG features are encoded into quantum states, and parameterized quantum gates are used for classification. The parameters are optimized using classical algorithms to improve performance.

Pro

This approach effectively captures complex nonlinear relationships in EEG data and improves classification accuracy. It also provides flexibility.

Limitations

The optimization process is computationally expensive and requires careful tuning. The performance also depends on the quality of quantum hardware, which may introduce noise and errors.

2.9 QUANTUM DEEP LEARNING FOR EEG SIGNAL CLASSIFICATION

Technique of Work Use

A quantum deep learning framework has been developed by integrating quantum layers into classical neural networks. Classical layers handle initial feature extraction, while quantum layers improve representation and learning capability. This hybrid setup utilizes the strengths of both computing approaches.

Pros

The model improves classification performance and enhances feature learning capability. It effectively handles complex EEG data and provides better generalization compared to traditional deep learning models.

Limitations

The model faces scalability issues due to limitations in quantum hardware and the number of available qubits. Training is also computationally intensive and requires advanced resources.

2.10 EEG-BASED BRAIN-COMPUTER INTERFACE ENABLES REAL-TIME ROBOTIC FINGER CONTROL

Technique of Work Use

This work develops a real-time EEG-based BCI system that decodes motor imagery

signals to control robotic finger movements. EEG signals are preprocessed, relevant features are extracted, and a classification model translates brain activity into robotic control commands.

Pros

It enables real-time brain-to-robot interaction and is useful for assistive and rehabilitation applications, especially for users with motor disabilities.

Limitations

Performance is affected by EEG noise, user variability, and the need for frequent calibration, limiting robustness in real-world environments.

2.11 RESEARCH GAPS

Despite rapid advancements in Brain–Computer Interface (BCI) systems and the integration of quantum machine learning techniques, several critical challenges remain unresolved in existing research.

Firstly, most traditional EEG-based BCI systems rely heavily on classical machine learning and deep learning approaches. These methods often struggle to effectively handle EEG signals due to their low signal-to-noise ratio, non-stationary behaviour, and high inter-subject variability, which leads to reduced classification accuracy and limited robustness in real-world scenarios.

Secondly, although recent studies have introduced quantum machine learning models such as Quantum Neural Networks (QNN), variational quantum circuits, and quantum neuromorphic models, these approaches are still in their early stages. Most of these works are theoretical or simulation-based, and their performance on real quantum hardware remains limited due to issues such as qubit constraints, noise in quantum operations, and hardware instability.

Thirdly, existing quantum-assisted models are often designed for specific and isolated BCI tasks, such as emotion recognition, autism detection, motor imagery classification, or robotic control. However, there is a lack of a generalized and unified quantum-assisted framework that can be effectively applied across multiple EEG-based BCI applications.

In addition, quantum models are not yet optimized for real-time processing requirements, especially in time-sensitive applications like robotic control and assistive communication systems. The computational overhead of quantum simulations further limits their deployment in practical systems.

Moreover, very few studies focus on end-to-end integration of quantum feature extraction, classical optimization, and real-time classification, which is essential for building practical BCI

systems. Most existing approaches treat these components separately rather than as a unified pipeline.

Therefore, there is a clear need for a unified, scalable, and real-time quantum-assisted machine learning framework that can effectively handle noisy and high-dimensional EEG signals, improve classification accuracy, support multiple BCI applications, and ensure practical deployment with reduced computational complexity.

CHAPTER 3

SYSTEM DESIGN

3.1 OVERVIEW

This section presents the architecture of the proposed framework, designed for efficient EEG-based brain–computer interface classification using classical and quantum-inspired approaches.

3.2 ARCHITECTURAL DESIGN

The proposed system starts by collecting EEG data from BCI systems. This data includes brain signal recordings along with their corresponding labels, such as eye state or brain activity. The collected data is then stored as a dataset for further processing.

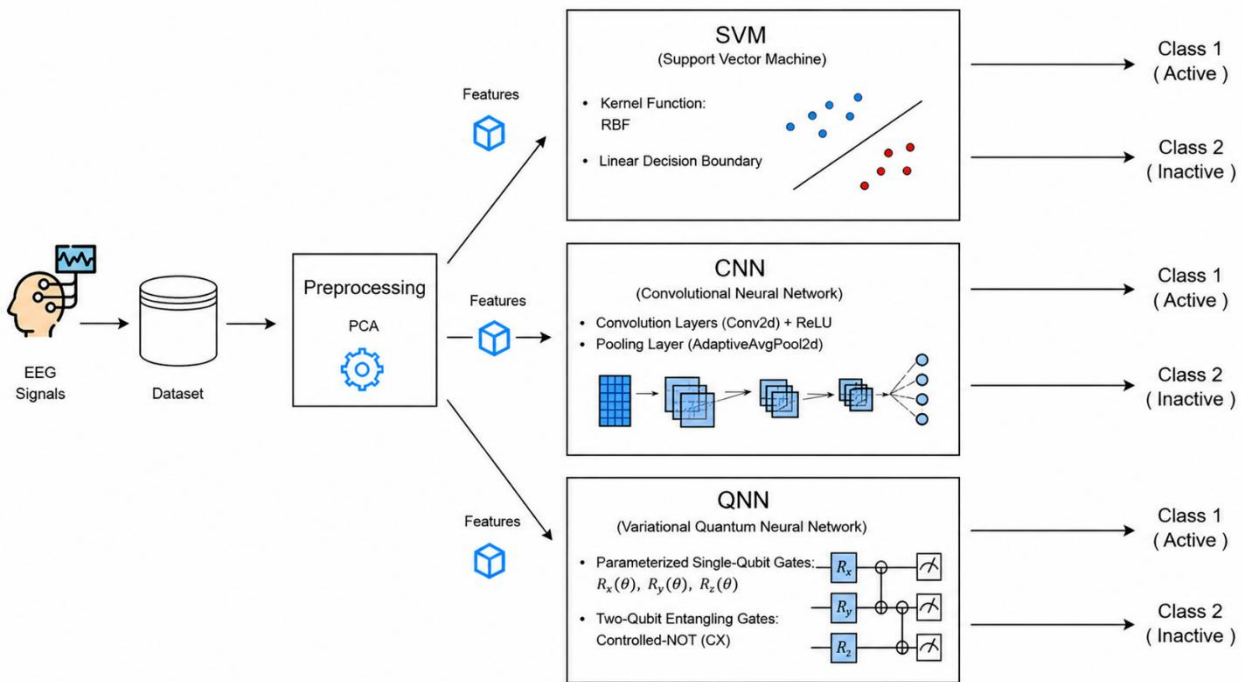


Figure 3.1 EEG Classification Architecture

In the next step, the data goes through preprocessing. Here, normalization is applied to bring all signal values to a common scale, making the data more consistent. After that, Principal Component Analysis (PCA) is used to reduce the number of features by keeping only the most important information, which helps remove noise and improves efficiency.

Once preprocessing is complete, the data is used for classification using three different models. First, the reduced features are given to the Support Vector Machine (SVM), which classifies the data by finding the best boundary between different classes.

Next, the same data is reshaped and passed into a Convolutional Neural Network (CNN). The CNN automatically learns patterns from the data by capturing spatial relationships, which reduces the need for manual feature extraction.

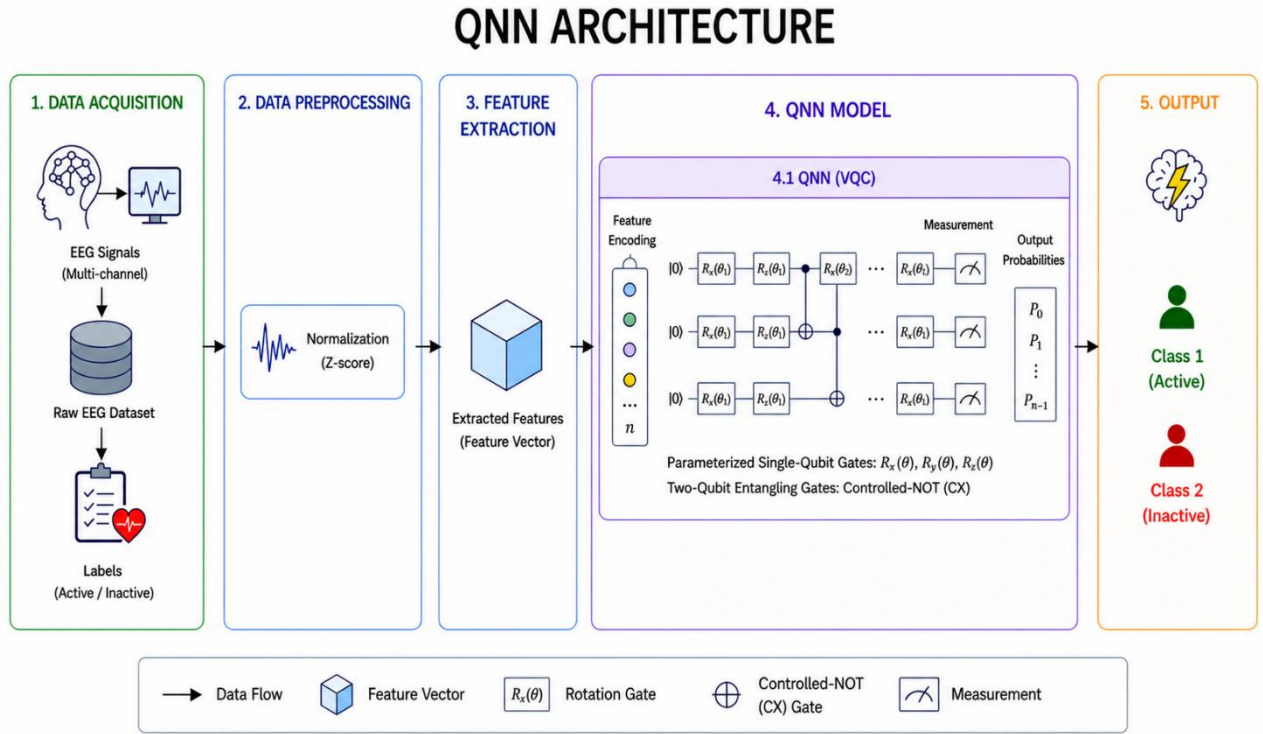


Figure 3.2 QNN Model Architecture

In addition, a Variational Quantum Neural Network (VQNN) is used. In this model, the input data is encoded into quantum states using parameterized quantum circuits. These circuits help in capturing complex and nonlinear relationships in EEG signals, which are difficult for classical models to learn.

Finally, the performance of all three models—SVM, CNN, and VQNN—is evaluated and compared. This comparison helps in understanding which model works better for EEG signal classification.

Overall, the proposed architecture improves the classification process, enhances feature learning, and provides a clear comparison between classical and quantum-inspired approaches for BCI systems.

3.3 DATASET DESCRIPTION

Datasets play a key role in EEG-based brain–computer interface (BCI) systems because they help in understanding different patterns of brain activity from recorded signals. In this work, three EEG datasets are used to develop and evaluate the proposed QSVM–QNN framework.

The EEG Eye State Classification dataset comes with labelled EEG signals that represent two conditions: open eyes and closed eyes. Because the data is already labelled, it can be directly used for training and testing the model without any additional labelling process.

Dataset	#Samples	#Features
EEG Eye State Classification Dataset	14,980	14

Table 6.1 Details of dataset

Column	Brain Region	What it Represents
AF3	Anterior frontal (left)	Attention, focus
AF4	Anterior frontal (right)	Attention, decision making
F3	Frontal (left)	Thinking, memory
F4	Frontal (right)	Cognitive control
F7	Frontal lateral (left)	Emotion processing
F8	Frontal lateral (right)	Emotion processing
FC5	Fronto-central (left)	Motor planning
FC6	Fronto-central (right)	Motor planning
T7	Temporal (left)	Auditory processing
T8	Temporal (right)	Auditory processing
P7	Parietal (left)	Spatial awareness
P8	Parietal (right)	Spatial awareness
O1	Occipital (left)	Visual processing
O2	Occipital (right)	Visual processing

Table 6.2 Feature Columns in Eye State Dataset

The dataset provides a balanced representation of brain activity across multiple regions, enabling effective learning of spatial patterns in EEG signals. These features are well-suited for classification tasks, as they capture both cognitive and sensory responses associated with eye states.

3.4 EVALUATION METRICS

The performance of the proposed system is evaluated using standard classification metrics to ensure reliability and effectiveness.

Accuracy:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision:

$$Precision = \frac{TP}{TP + FP}$$

Recall:

$$Recall = \frac{TP}{TP + FN}$$

F1 Score:

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

3.5 LIST OF MODULES

- EEG DATA INPUT AND PREPROCESSING MODULE
- DIMENSIONALITY REDUCTION MODULE (PCA)
- QUANTUM NEURAL NETWORK (QNN) MODULE
- PERFORMANCE EVALUATION AND COMPARISON

3.5.1 EEG DATA INPUT AND PREPROCESSING MODULE

This module is responsible for loading and preparing the EEG dataset for further processing. The EEG Eye State dataset is used, where each sample contains multiple features representing brain signals and a label indicating the eye state (open or closed).

To improve the quality of the data, standardization is applied so that all features are brought to a similar scale. This helps the models learn more effectively and avoids bias toward larger values. The scaling process used in the code is:

$$X_{scaled} = \frac{X - \text{mean}(X)}{\text{std}(X)}$$

After scaling, the dataset is divided into training and validation sets. Since the dataset contains 14 features, padding is applied to convert it into 16 features, which are then reshaped into a 4×4 format. This reshaping is necessary for CNN and VQNN models.

3.5.2 DIMENSIONALITY REDUCTION MODULE (PCA)

This module reduces the number of features in the EEG dataset to make processing more efficient. Since EEG data contains many features, some of them may be redundant or add noise. To address this, Principal Component Analysis (PCA) is used to select only the most important information.

PCA transforms the original features into a smaller set of principal components that capture most of the data's variance. Instead of using all features, only these key components are retained.

In the implementation, PCA is applied as:

$$X_{PCA} = \text{PCA}(X)$$

The number of components is chosen to preserve 95% of the total variance. This helps in reducing noise, lowering computational cost, and improving model performance.

The reduced dataset is then used for SVM classification, making the model faster and more effective. Overall, this module simplifies the data while keeping its essential information intact.

3.5.3 QUANTUM NEURAL NETWORK (QNN) MODULE

In This module implements a Variational Quantum Neural Network (VQNN) to learn complex patterns from EEG data using a hybrid quantum–classical approach. The input features are first processed through a classical layer and then passed into a quantum circuit.

Each input value is encoded into qubits using rotation gates:

$$RY(x_i)$$

Entanglement between qubits is created using CNOT gates to capture relationships between features. Trainable parameters are applied using:

$$RY(\theta_i)$$

These parameters are updated during training to improve learning.

The output of the quantum circuit is obtained by measuring expectation values:

$$y = \langle Z \rangle$$

These outputs are then passed through a final layer for classification. The model is trained using the loss function:

$$L = \text{CrossEntropy}(y_{pred}, y_{true})$$

This module helps the system learn complex and nonlinear patterns in EEG signals more effectively.

3.5.4 PERFORMANCE EVALUATION AND COMPARISON

This module is used to measure how well the implemented models—SVM, CNN, and VQNN—perform in classifying EEG signals. The primary metric used for evaluation is accuracy, which indicates how many predictions made by the model are correct out of the total number of samples.

The accuracy is calculated as:

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Samples}}$$

In the implementation, accuracy is obtained by comparing the predicted labels with the actual labels in the dataset. In addition to accuracy, other important evaluation metrics such as precision, recall, and F1-score are also used to provide a more detailed performance analysis.

Each model is trained and tested using the same dataset and conditions to ensure a fair comparison. After training, the performance of all three models is evaluated, and their accuracy values are compared.

This comparison helps in identifying which model performs better for EEG signal classification. It also provides useful insights into how traditional approaches like SVM and CNN differ from quantum-based methods like VQNN in terms of learning capability and overall performance.

3.6 SUMMARY

This section explains the overall workflow of the proposed EEG signal classification system. The process begins with the preprocessing module, where raw EEG data is cleaned and standardized to improve its quality and consistency. After that, Principal Component Analysis (PCA) is applied to reduce the number of features while retaining the most important information, which helps simplify the data and improve efficiency.

For classification, three different models are used. The SVM model works on PCA-reduced features for efficient classification, the CNN model automatically learns patterns from structured EEG data, and the VQNN model captures complex nonlinear relationships using quantum-based techniques. Finally, the performance of all models is evaluated using accuracy, and the results are compared to determine the most effective approach for EEG signal classification.

CHAPTER 4

SYSTEM IMPLEMENTATION

4.1 PRINCIPAL COMPONENT ANALYSIS (PCA)

The EEG dataset contains high-dimensional features that often include redundant and correlated information. To improve efficiency and reduce complexity, PCA is used as a dimensionality reduction technique. First, the data is standardized so that all features have equal scale with zero mean and unit variance. PCA then identifies the directions of maximum variance in the data and transforms the original features into a new set of uncorrelated components.

From these components, only the most important ones are selected such that about 95% of the total information is retained. The original data is then projected into this reduced feature space. Thus, PCA reduces the number of features while preserving essential information, improving both computational efficiency and model performance.

4.2 SUPPORT VECTOR MACHINE (SVM)

After dimensionality reduction using PCA, the processed features are used as input to a Support Vector Machine (SVM) classifier for final prediction. SVM is a supervised learning algorithm that finds an optimal hyperplane to separate different classes with maximum margin. In this work, a non-linear RBF (Radial Basis Function) kernel is used to handle complex relationships in EEG data.

The model is trained using the PCA-transformed training data, and hyperparameters such as the regularization parameter (C) and kernel coefficient (γ) are tuned to improve performance. After training, the model predicts the class labels for the test data. Finally, the performance of the classifier is evaluated using accuracy, which measures the proportion of correctly classified samples.

4.3 CONVOLUTIONAL NEURAL NETWORK (CNN)

In this work, a Convolutional Neural Network (CNN) is used for EEG eye-state classification to automatically learn discriminative patterns from the input data. The EEG dataset is first standardized to ensure all features are on a uniform scale, improving model stability and convergence. Since CNN requires spatial structure as input, the 1D feature

vectors are padded and reshaped into a 2D format of size 4×4 , allowing the data to be processed as image-like inputs.

The convolution operation in CNN extracts local patterns from the input feature map. It is defined as:

$$F(x) = (X * W) + b$$

where X is the input, W is the filter (kernel), and b is the bias term.

After convolution, a non-linear activation function is applied: $ReLU(x) = \max(0, x)$ to introduce non-linearity and improve learning capability.

The network also uses adaptive average pooling to reduce spatial dimensions while preserving important information:

$$Pooled\ Feature = AvgPool(X)$$

The extracted feature maps are then flattened into a 1D vector and passed through fully connected layers, where the final classification is performed.

The output layer uses the Softmax function to generate class probabilities:

$$Softmax(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

For training, the Cross-Entropy Loss function is used:

$$L = -\sum y_i \log(\hat{y}_i)$$

where y_i is the true label and $\hat{y}_i$ is the predicted probability.

The model is optimized using the Adam optimizer, which updates parameters using adaptive learning rates based on first and second moments of gradients. Overall, CNN effectively learns hierarchical feature representations from EEG signals, improving classification accuracy without manual feature engineering.

4.4 QUANTUM NEURAL NETWORK

The proposed system implements a Variational Quantum Neural Network (QNN) for EEG-based eye-state classification. It follows a hybrid quantum–classical architecture, where classical EEG signals are first pre-processed and then encoded into a quantum circuit. The quantum circuit performs feature transformation using parameterized quantum gates, and the resulting quantum features are passed to a classical neural network for final

classification. The implementation is developed using Python, with PennyLane for quantum simulation and PyTorch for classical deep learning.

The EEG dataset consists of multiple signal features representing brain activity. Each sample is represented as:

$$x = [x_1, x_2, \dots, x_n]$$

To improve convergence and remove bias due to varying feature ranges, standardization is applied:

$$x' = \frac{x - \mu}{\sigma}$$

After preprocessing, the feature vector is padded and reshaped into a 4-qubit compatible format (16 features) for quantum processing.

Classical EEG features are encoded into quantum states using rotation gates. Each feature is mapped to a qubit using an RY rotation, which transforms classical data into quantum amplitudes.

The encoded quantum state is represented as:

$$|\psi(x)\rangle = \prod RY(x_i) |0\rangle$$

This step allows the quantum circuit to represent EEG signals in a higher-dimensional feature space.

To capture relationships between EEG features, entanglement is introduced using CNOT gates between adjacent qubits. This helps the model learn correlations between input features.

Trainable quantum parameters are applied using parameterized rotation gates:

$$U(\theta) = \prod RY(\theta_i)$$

The quantum circuit output is obtained after applying both entanglement and variational layers.

After quantum processing, the circuit is measured using expectation values of Pauli-Z operators. These measurements convert quantum states back into classical information.

The resulting feature vector is:

$$f(x) = [z_1, z_2, \dots, z_n]$$

This vector represents the quantum-enhanced features extracted from EEG signals.

The extracted quantum features are passed into a fully connected neural network for classification.

Linear Transformation

$$y = Wf(x) + b$$

A softmax function is applied to generate class probabilities for eye-state detection (open or closed).

The model is trained using cross-entropy loss, which measures the difference between predicted and actual labels:

$$\mathcal{L} = -\sum y \log(\hat{y})$$

The parameters of both quantum and classical layers are optimized using gradient-based optimization (Adam optimizer).

4.5 Summary

The system implementation involves preprocessing EEG data, extracting relevant features, and applying three models—SVM, CNN, and VQNN—for classification. Each model is trained and tested under the same conditions to ensure consistency. The implementation integrates classical and quantum approaches to analyze their effectiveness in EEG signal classification.

CHAPTER 5

RESULTS AND DISCUSSION

5.1. SVM Model Performance

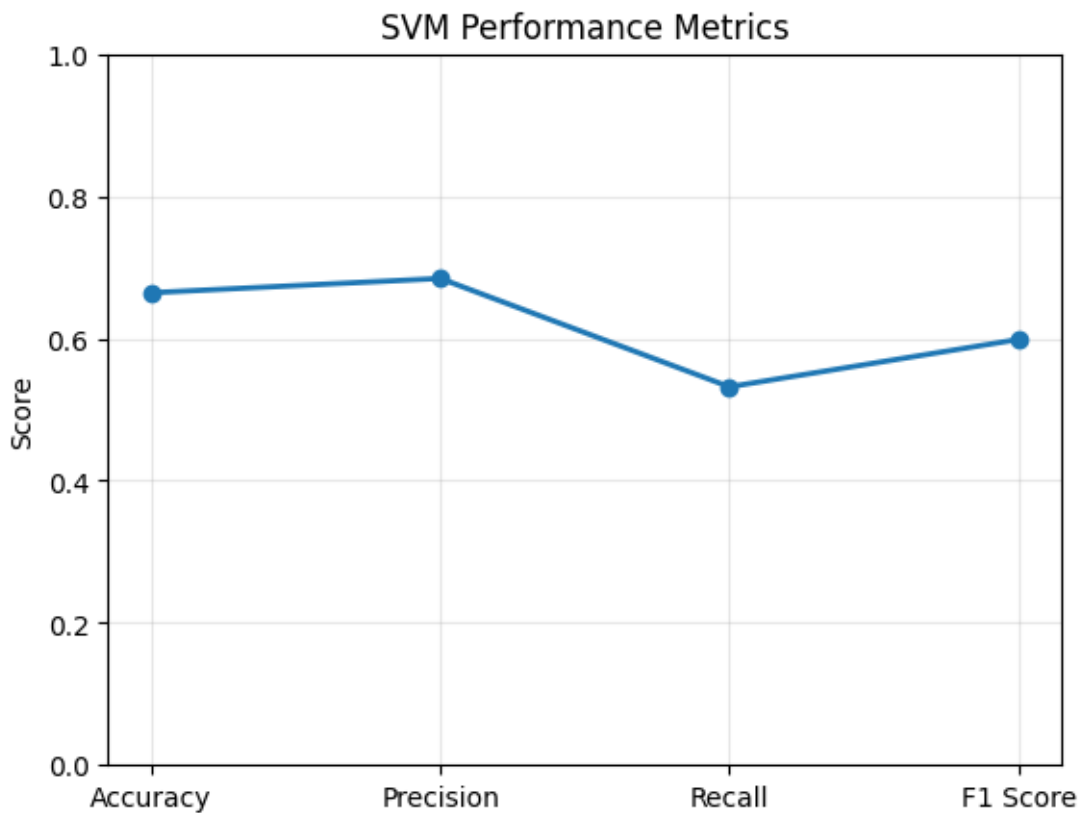


Fig.6.1 SVM Performance Metrics

The performance of the Support Vector Machine (SVM) model is analyzed using Accuracy, Precision, Recall, and F1-score.

- The model achieves moderate accuracy, indicating acceptable overall classification performance.
- Precision is relatively stable, showing that predicted eye-state labels are reasonably correct.
- Recall is slightly lower, suggesting that some actual eye-state instances are not correctly identified.
- The F1-score reflects a balanced but limited performance.

Overall, SVM provides a strong baseline but struggles to capture complex patterns in EEG signals due to limited feature learning capability.

5.2. CNN Model Performance

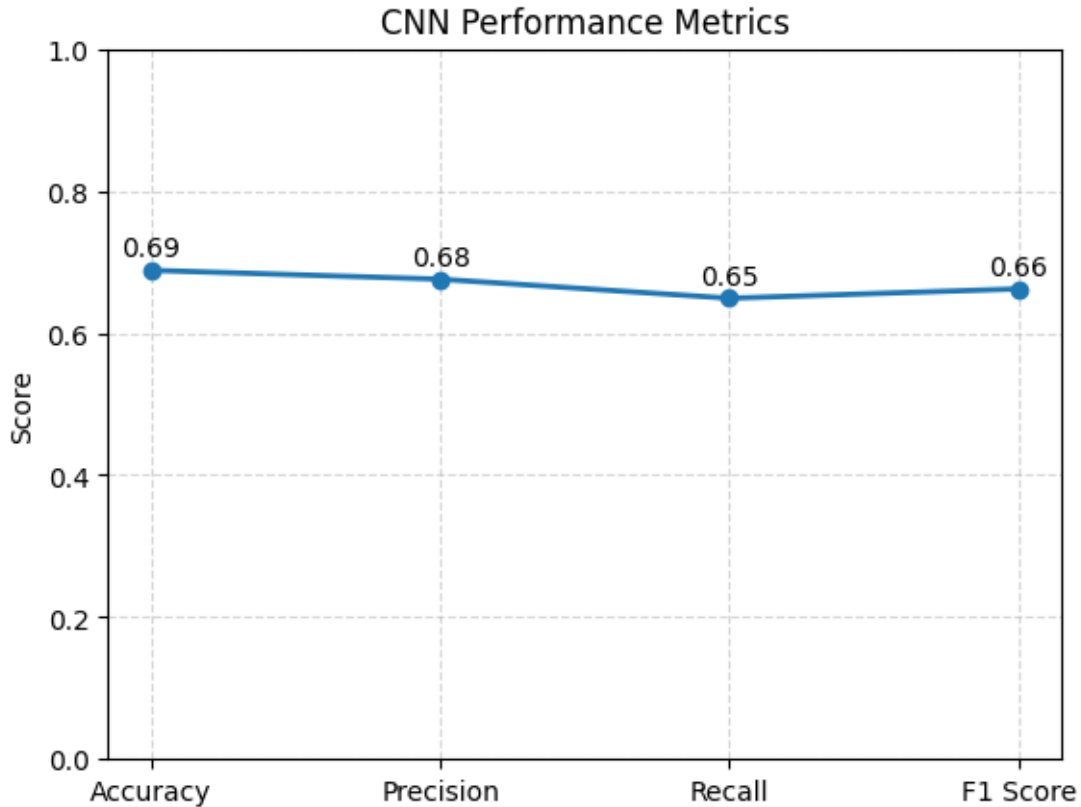


Fig.6.2 SVM Performance Metrics

The Convolutional Neural Network (CNN) demonstrates improved performance compared to SVM.

- Accuracy is higher, indicating better overall classification capability.
- Precision improves due to effective feature extraction.
- Recall is also higher, meaning the model detects more true eye-state cases.
- The F1-score shows a balanced improvement across both precision and recall.

CNN performs well because it automatically extracts meaningful spatial features from EEG data, reducing the need for manual feature engineering.

5.3. QNN Model Performance

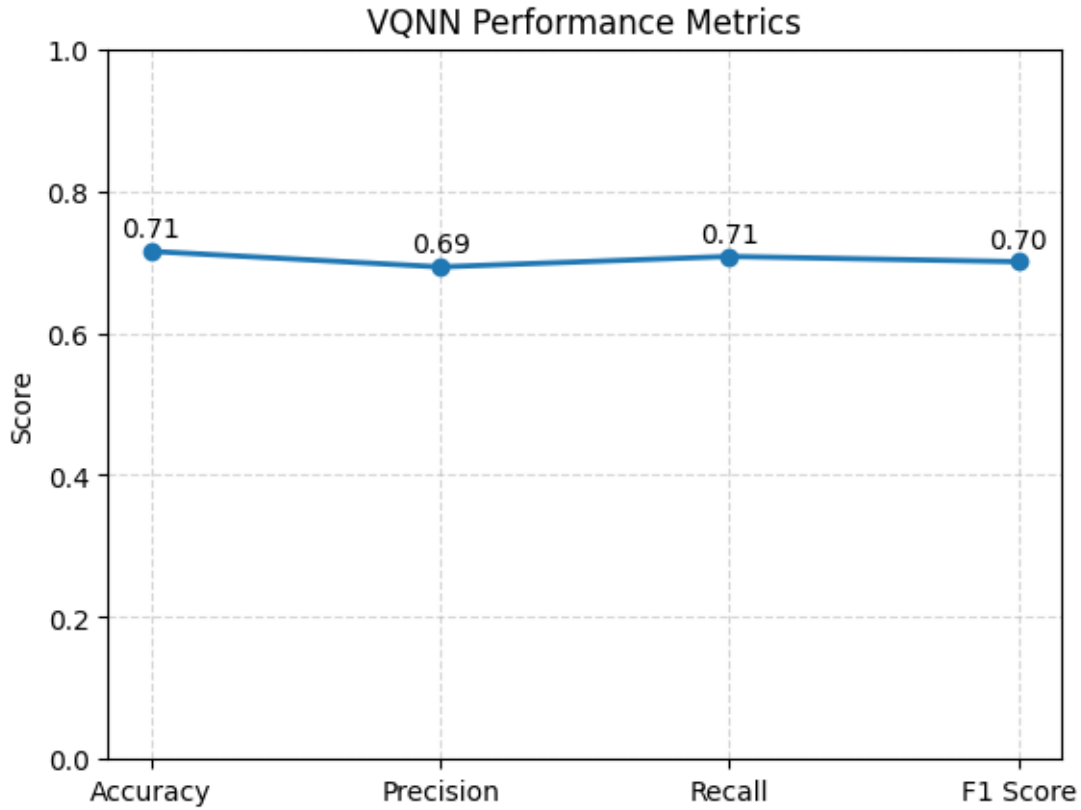


Fig.6.3 QNN Performance Metrics

The proposed Variational Quantum Neural Network (QNN) achieves the best performance among all models.

- It records the highest accuracy, indicating superior classification performance.
- Precision is high, meaning predictions are highly reliable.
- Recall is also high, showing the model effectively captures true eye-state instances.
- The F1-score confirms a strong balance between precision and recall.

The improved performance is due to quantum feature encoding and entanglement, which enhance the model's ability to capture complex nonlinear relationships in EEG signals.

5.4. Comparative Analysis of Models

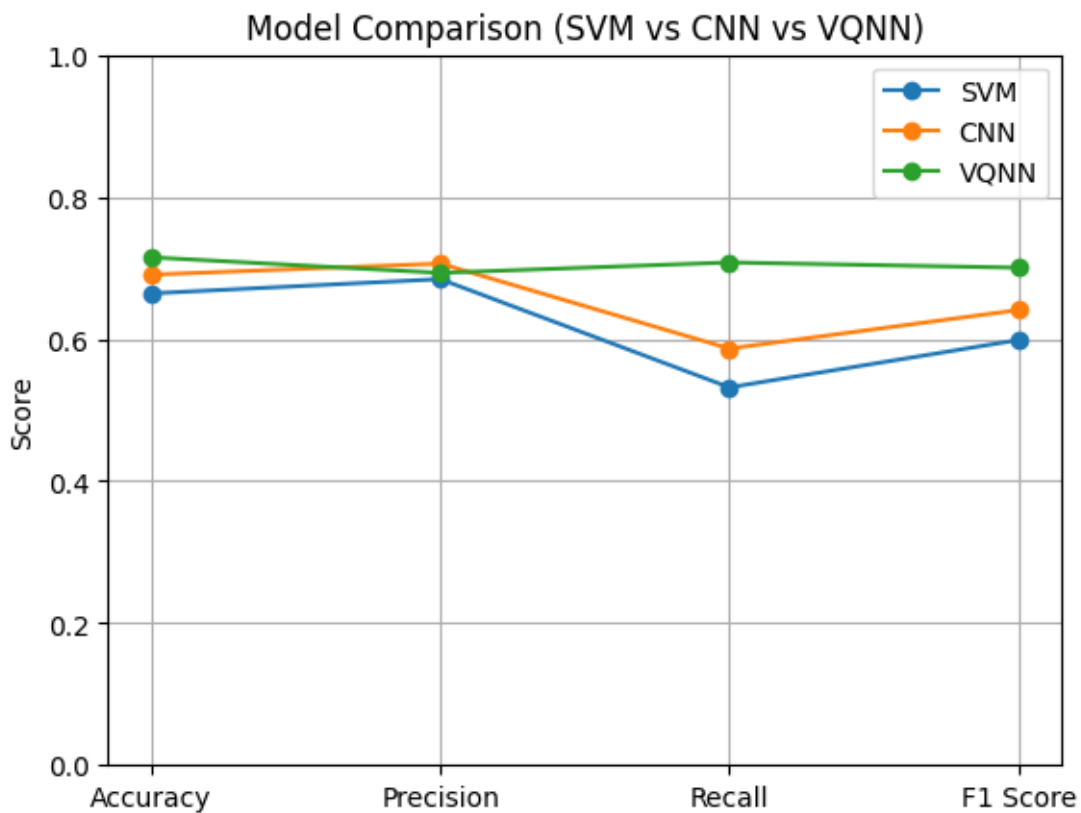


Fig.6.4 Models Comparison

The comparative graph illustrates the performance differences between SVM, CNN, and QNN across all evaluation metrics.

Key Observations:

- Accuracy: QNN achieves the highest accuracy, followed by CNN and SVM
- Precision: QNN provides more reliable predictions with fewer false positives
- Recall: QNN detects more actual eye-state cases compared to other models
- F1-Score: QNN maintains the best balance between precision and recall

Performance Ranking:

QNN > CNN > SVM

5.5. Discussion

The results clearly show that:

- SVM is limited by its inability to automatically learn complex features

- CNN improves performance through deep feature extraction
- QNN further enhances classification by leveraging quantum computation principles

The QNN model's ability to map data into a higher-dimensional quantum space allows it to better represent EEG signal patterns, leading to improved performance across all metrics.

5.6. Summary

The results show the comparative performance of SVM, CNN, and VQNN using metrics such as accuracy, precision, recall, and F1-score. The analysis highlights the strengths and limitations of each model, with VQNN demonstrating improved capability in handling complex patterns. The discussion provides insights into the effectiveness of quantum-based methods compared to traditional techniques for EEG classification.

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENT(S)

6.1 CONCLUSION

In Brain–Computer Interface (BCI) systems, EEG signals are analyzed to understand and interpret brain activity, but traditional machine learning methods often face limitations such as lower accuracy, sensitivity to noise, and difficulty in handling complex and high-dimensional data. To overcome these challenges, this project adopts a comparative approach using three models: Support Vector Machine (SVM), Convolutional Neural Network (CNN), and Variational Quantum Neural Network (VQNN) for effective EEG signal classification.

The system begins with preprocessing steps such as normalization to standardize the data and Principal Component Analysis (PCA) to reduce dimensionality and retain the most important features. The SVM model uses these optimized features for classification, while the CNN model automatically learns spatial patterns from reshaped EEG data, reducing the need for manual feature extraction. In addition, the VQNN model applies quantum-inspired techniques using parameterized quantum circuits to capture complex nonlinear relationships in EEG signals. By comparing these models, the study highlights the strengths of both classical and quantum-based approaches.

The experimental results show that all models provide reliable performance, and the overall framework offers an efficient and scalable solution for EEG signal classification in BCI applications.

6.2 FUTURE ENHANCEMENT

Future work can focus on improving the overall performance of the system by fine-tuning model parameters and enhancing feature representation techniques to achieve higher classification accuracy. The study can also be extended to larger and more diverse EEG datasets to better evaluate the models under different conditions and improve their generalization capability across different users and environments.

In addition, future research may explore implementing quantum-based models such as VQNN on real quantum hardware to assess their practical feasibility and performance in real-world scenarios. Advanced preprocessing techniques, including improved dimensionality reduction and feature extraction methods, can also be incorporated to enhance efficiency and reduce computational complexity.

Further improvements may include integrating more advanced deep learning and hybrid quantum-classical models to strengthen learning capability and robustness. Efforts can also be made to develop real-time EEG signal processing systems and user-friendly visualization tools, making the framework more practical, accessible, and suitable for real-world Brain–Computer Interface applications.

APPENDIX 1

SYSTEM REQUIREMENT

Hardware Requirement

Processor	:	Intel Core i5 (or higher)
RAM	:	8 GB minimum
GPU (Optional)	:	NVIDIA GPU with CUDA support

Software Requirement

Operating System	:	WINDOWS
Language used	:	Python
Platform	:	Google Colab

APPENDIX II CODING

SVM:

```

import pandas as pd
import numpy as np
from sklearn.metrics import accuracy_score
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC

from sklearn.model_selection import GridSearchCV
from sklearn.decomposition import PCA
df = pd.read_csv("EEG_Eye_State_Classification.csv")
print(df.shape)
X = df.drop('eyeDetection', axis=1)
y = df['eyeDetection']

print(df["eyeDetection"].unique())
print(df["eyeDetection"].dtype)

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
#SVM

param_grid = [
    {'kernel': ['linear'], 'C': [1, 10]},
    {'kernel': ['rbf'], 'C': [1, 10], 'gamma': ['scale', 0.01]}
]
```

```
grid = GridSearchCV(SVC(), param_grid, cv=3, scoring='accuracy', n_jobs=-1)
grid.fit(X_train, y_train)

print("Best Parameters:", grid.best_params_)
print("Best Accuracy:", grid.best_score_)

best_model = grid.best_estimator_
y_pred = best_model.predict(X_test)

print("Test Accuracy:", accuracy_score(y_test, y_pred))
pca = PCA(n_components=0.95)

X_train_pca = pca.fit_transform(X_train)
X_test_pca = pca.transform(X_test)

print(pca.n_components_)
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

model = SVC(kernel='rbf', C=10, gamma=0.01)
model.fit(X_train_pca, y_train)

y_pred = model.predict(X_test_pca)

# Metrics
acc = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred, average='binary')
recall = recall_score(y_test, y_pred, average='binary')
f1 = f1_score(y_test, y_pred, average='binary')

print("SVM Accuracy:", acc)
print("Precision:", precision)
print("Recall:", recall)
```

```

print("F1 Score:", f1)
import matplotlib.pyplot as plt

x = [0, 1, 2, 3]
labels = ['Accuracy', 'Precision', 'Recall', 'F1 Score']
values = [acc, precision, recall, f1]

plt.figure()

plt.plot(x, values, marker='o', linestyle='-', linewidth=2)
plt.xticks(x, labels)

plt.title("SVM Performance Metrics")
plt.ylim(0, 1)
plt.ylabel("Score")
plt.grid(True, alpha=0.3)

plt.show()

```

CNN:

```

# =====
# IMPORTS
# =====

import pandas as pd
import numpy as np
import torch
import torch.nn as nn
from torch.utils.data import DataLoader, TensorDataset
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

```

```

# =====
# DEVICE
# =====

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")

# =====
# LOAD DATA
# =====

df = pd.read_csv("EEG_Eye_State_Classification.csv")

X = df.drop("eyeDetection", axis=1).values
y = df["eyeDetection"].values

# =====
# PREPROCESSING
# =====

scaler = StandardScaler()
X = scaler.fit_transform(X)

X_train, X_val, y_train, y_val = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# pad 14 → 16 features
X_train = np.pad(X_train, ((0,0),(0,2)))
X_val = np.pad(X_val, ((0,0),(0,2)))

# reshape → CNN input (N, C, H, W)
X_train = X_train.reshape(-1, 1, 4, 4)
X_val = X_val.reshape(-1, 1, 4, 4)

# =====
# TENSORS

```

```

# =====
X_train = torch.tensor(X_train, dtype=torch.float32)
X_val = torch.tensor(X_val, dtype=torch.float32)

y_train = torch.tensor(y_train, dtype=torch.long)
y_val = torch.tensor(y_val, dtype=torch.long)

train_loader = DataLoader(TensorDataset(X_train, y_train), batch_size=16, shuffle=True)
val_loader = DataLoader(TensorDataset(X_val, y_val), batch_size=16)

# =====
# CNN MODEL (FIXED)
# =====
class EEG_CNN(nn.Module):
    def __init__(self):
        super(EEG_CNN, self).__init__()

        self.conv = nn.Sequential(
            nn.Conv2d(1, 16, kernel_size=2),
            nn.ReLU(),

            nn.Conv2d(16, 32, kernel_size=2, padding=1),
            nn.ReLU()
        )

        self.pool = nn.AdaptiveAvgPool2d((2, 2))

        self.fc = nn.Sequential(
            nn.Linear(32 * 2 * 2, 64),
            nn.ReLU(),
            nn.Linear(64, 2)
        )

```

```
def forward(self, x):
```

```
    x = self.conv(x)
```

```
    x = self.pool(x)
```

```
    x = x.view(x.size(0), -1)
```

```
    x = self.fc(x)
```

```
    return x
```

```
model = EEG_CNN().to(device)
```

```
# =====
```

```
# LOSS & OPTIMIZER
```

```
# =====
```

```
criterion = nn.CrossEntropyLoss()
```

```
optimizer = torch.optim.Adam(model.parameters(), lr=0.001)
```

```
# =====
```

```
# TRAIN FUNCTION
```

```
# =====
```

```
def train_cnn(model, train_loader, val_loader, epochs=15):
```

```
    for epoch in range(epochs):
```

```
        # TRAIN
```

```
        model.train()
```

```
        total_loss = 0
```

```
        correct = 0
```

```
        for X_batch, y_batch in train_loader:
```

```
            X_batch, y_batch = X_batch.to(device), y_batch.to(device)
```

```
            optimizer.zero_grad()
```

```
            outputs = model(X_batch)
```

```
            loss = criterion(outputs, y_batch)
```

```

    loss.backward()
    optimizer.step()

    total_loss += loss.item()
    preds = torch.argmax(outputs, dim=1)
    correct += (preds == y_batch).sum().item()

train_acc = correct / len(train_loader.dataset)

# VALIDATION
model.eval()
correct = 0

with torch.no_grad():
    for X_batch, y_batch in val_loader:
        X_batch, y_batch = X_batch.to(device), y_batch.to(device)

        outputs = model(X_batch)
        preds = torch.argmax(outputs, dim=1)
        correct += (preds == y_batch).sum().item()

val_acc = correct / len(val_loader.dataset)

print(f'Epoch {epoch+1:02d} | Loss: {total_loss:.4f} | '
      f'Train Acc: {train_acc:.4f} | Val Acc: {val_acc:.4f}')

# =====
# RUN
# =====

train_cnn(model, train_loader, val_loader, epochs=5)

from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

```

```

def evaluate_metrics(model, loader):
    model.eval()

    y_true = []
    y_pred = []

    with torch.no_grad():
        for X_batch, y_batch in loader:
            X_batch = X_batch.to(device)

            outputs = model(X_batch)
            preds = torch.argmax(outputs, dim=1).cpu().numpy()

            y_pred.extend(preds)
            y_true.extend(y_batch.numpy())

    acc = accuracy_score(y_true, y_pred)
    precision = precision_score(y_true, y_pred)
    recall = recall_score(y_true, y_pred)
    f1 = f1_score(y_true, y_pred)

    return acc, precision, recall, f1

val_acc, val_prec, val_recall, val_f1 = evaluate_metrics(model, val_loader)

print("\n===== VALIDATION METRICS =====")
print(f'Accuracy : {val_acc:.4f}')
print(f'Precision : {val_prec:.4f}')
print(f'Recall : {val_recall:.4f}')
print(f'F1 Score : {val_f1:.4f}')

import matplotlib.pyplot as plt

```



```

# Labels and values
metrics = ['Accuracy', 'Precision', 'Recall', 'F1 Score']
values = [val_acc, val_prec, val_recall, val_f1]

plt.figure()

plt.plot(metrics, values, marker='o', linewidth=2)

plt.title("CNN Performance Metrics")
plt.ylim(0, 1)
plt.ylabel("Score")

# Add value labels
for i, v in enumerate(values):
    plt.text(i, v + 0.02, f'{v:.2f}', ha='center')

plt.grid(True, linestyle='--', alpha=0.5)
plt.show()

```

QNN:

```

#Variational Quantum Neural Network (VQNN)

# =====
# IMPORTS
# =====

import pandas as pd
import numpy as np
import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import DataLoader, TensorDataset

```

```

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

import pennylane as qml

# Device
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")

# =====
# LOAD DATA
# =====
df = pd.read_csv("EEG_Eye_State_Classification.csv")

X = df.drop("eyeDetection", axis=1).values
y = df["eyeDetection"].values

# =====
# PREPROCESSING
# =====
scaler = StandardScaler()
X = scaler.fit_transform(X)

# Split
X_train, X_val, y_train, y_val = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Convert to 4x4 (pad 14 → 16)
X_train = np.pad(X_train, ((0,0),(0,2)))
X_val = np.pad(X_val, ((0,0),(0,2)))

X_train = X_train.reshape(-1, 1, 4, 4)
X_val = X_val.reshape(-1, 1, 4, 4)

```

```

# =====
# TENSORS
# =====

X_train = torch.tensor(X_train, dtype=torch.float32)
X_val = torch.tensor(X_val, dtype=torch.float32)

y_train = torch.tensor(y_train, dtype=torch.long)
y_val = torch.tensor(y_val, dtype=torch.long)

train_loader = DataLoader(TensorDataset(X_train, y_train), batch_size=16, shuffle=True)
val_loader = DataLoader(TensorDataset(X_val, y_val), batch_size=16)

# =====
# QUANTUM CIRCUIT
# =====

n_qubits = 4
dev = qml.device("default.qubit", wires=n_qubits)

@qml.qnode(dev, interface="torch")
def quantum_net(inputs, weights):
    # Encode data
    for i in range(n_qubits):
        qml.RY(inputs[i], wires=i)

    # Entanglement
    for i in range(n_qubits - 1):
        qml.CNOT(wires=[i, i+1])

    # Trainable layer
    for i in range(n_qubits):
        qml.RY(weights[i], wires=i)

```

```
return [qml.expval(qml.PauliZ(i)) for i in range(n_qubits)]
```

```
# =====
```

```
# HYBRID MODEL
```

```
# =====
```

```
class HybridModel(nn.Module):
```

```
    def __init__(self):
```

```
        super().__init__()
```

```
        self.flatten = nn.Flatten()
```

```
        self.fc1 = nn.Linear(16, 4)
```

```
        self.q_weights = nn.Parameter(0.01 * torch.randn(4))
```

```
        self.fc2 = nn.Linear(4, 2)
```

```
    def forward(self, x):
```

```
        x = self.flatten(x)
```

```
        x = torch.tanh(self.fc1(x))
```

```
        q_out = []
```

```
        for i in range(x.shape[0]):
```

```
            # FIX: keep dtype consistent + preserve gradients
```

```
            q_result = quantum_net(x[i], self.q_weights)
```

```
            q_result = torch.stack(q_result).float()
```

```
            q_out.append(q_result)
```

```
        q_out = torch.stack(q_out).to(device)
```

```
        x = self.fc2(q_out)
```

```
        return x
```

```
# =====
```

```
# MODEL SETUP
```

```

# =====

model = HybridModel().to(device)

criterion = nn.CrossEntropyLoss()
optimizer = optim.Adam(model.parameters(), lr=0.01)

# =====
# TRAINING
# =====

def train_model(model, epochs=15):
    for epoch in range(epochs):
        model.train()
        total_loss = 0
        correct = 0

        for X_batch, y_batch in train_loader:
            X_batch, y_batch = X_batch.to(device), y_batch.to(device)

            optimizer.zero_grad()
            outputs = model(X_batch)

            loss = criterion(outputs, y_batch)
            loss.backward()
            optimizer.step()

            total_loss += loss.item()
            preds = torch.argmax(outputs, dim=1)
            correct += (preds == y_batch).sum().item()

        acc = correct / len(train_loader.dataset)
        print(f'Epoch {epoch+1}: Loss={total_loss:.4f}, Accuracy={acc:.4f}')

# =====

```

```

# TRAIN

# =====

train_model(model, epochs=20)

from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

# =====

# EVALUATION

# =====

model.eval()

all_preds = []
all_labels = []

with torch.no_grad():
    for X_batch, y_batch in val_loader:
        X_batch, y_batch = X_batch.to(device), y_batch.to(device)

        outputs = model(X_batch)
        preds = torch.argmax(outputs, dim=1)

        all_preds.extend(preds.cpu().numpy())
        all_labels.extend(y_batch.cpu().numpy())

# =====

# METRICS

# =====

accuracy = accuracy_score(all_labels, all_preds)
precision = precision_score(all_labels, all_preds, average='binary')
recall = recall_score(all_labels, all_preds, average='binary')
f1 = f1_score(all_labels, all_preds, average='binary')

print("\n===== VQNN PERFORMANCE =====")

```

```

print("Accuracy :", accuracy)
print("Precision :", precision)
print("Recall   :", recall)
print("F1 Score :", f1)

import matplotlib.pyplot as plt

# Metrics
metrics = ['Accuracy', 'Precision', 'Recall', 'F1 Score']
values = [accuracy, precision, recall, f1]

plt.figure()

plt.plot(metrics, values, marker='o', linewidth=2)

plt.title("VQNN Performance Metrics")
plt.ylim(0, 1)
plt.ylabel("Score")

# Add value labels
for i, v in enumerate(values):
    plt.text(i, v + 0.02, f'{v:.2f}', ha='center')

plt.grid(True, linestyle='--', alpha=0.5)
plt.show()

```

Comparison:

```

svm_metrics = [0.6642, 0.6843, 0.5319, 0.5986]
cnn_metrics = [0.6906, 0.7062, 0.5865, 0.6408]
vqnn_metrics = [0.7150, 0.6931, 0.7078, 0.7004]

labels = ['Accuracy', 'Precision', 'Recall', 'F1 Score']

```

```
plt.figure()

plt.plot(labels, svm_metrics, marker='o', label='SVM')
plt.plot(labels, cnn_metrics, marker='o', label='CNN')
plt.plot(labels, vqnn_metrics, marker='o', label='VQNN')

plt.title("Model Comparison (SVM vs CNN vs VQNN)")
plt.ylim(0, 1)
plt.ylabel("Score")
plt.legend()
plt.grid(True)

plt.show()
```

```
import matplotlib.pyplot as plt
```

```
# Define values FIRST
```

```
models = ['SVM', 'CNN', 'VQNN']
```

```
accuracies = [0.6642, 0.6906, 0.7150]
```

```
plt.figure()
```

```
plt.plot(models, accuracies, marker='o')
```

```
plt.title("Accuracy Comparison")
```

```
plt.ylim(0, 1)
```

```
plt.ylabel("Accuracy")
```

```
plt.grid(True)
```

```
plt.show()
```


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